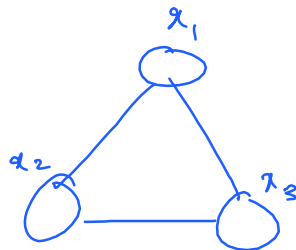


## Chapter 13

# Factor Graphs and Sum/Max-product Algorithms \*\*

This MRF implies the factorization



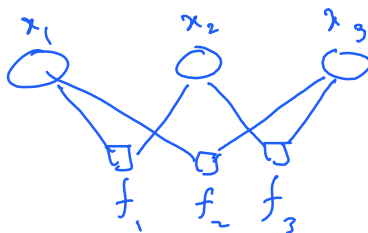
$$p(x_1, x_2, x_3) \propto \psi(x_1, x_2, x_3)$$

But suppose we actually want to represent

$$p(x_1, x_2, x_3) \propto f(x_1, x_2) f(x_2, x_3) f(x_3, x_1)$$

Is there a way to do this with a graph?

Factor graph:



Two types of nodes:

$V$ : variables:  $x_1, x_2, x_3$

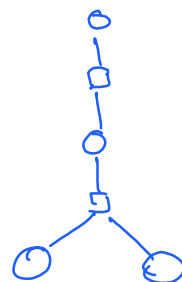
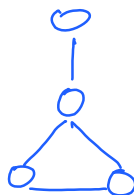
$F$ : factors:  $f_1, f_2, f_3$

$$p(x_1^m) = \prod_{f_i \in F} f_i(x_{f_i})$$

variable nodes adjacent to  $f_i$

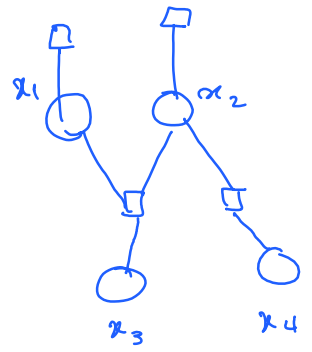
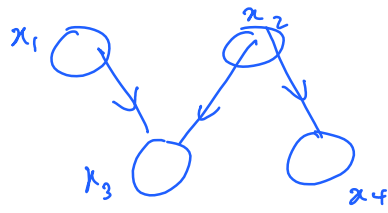
**MRF  $\rightarrow$  FG**

nodes  $\rightarrow$  variable nodes  
maximal cliques  $\rightarrow$  factor nodes



**BN  $\rightarrow$  FG**

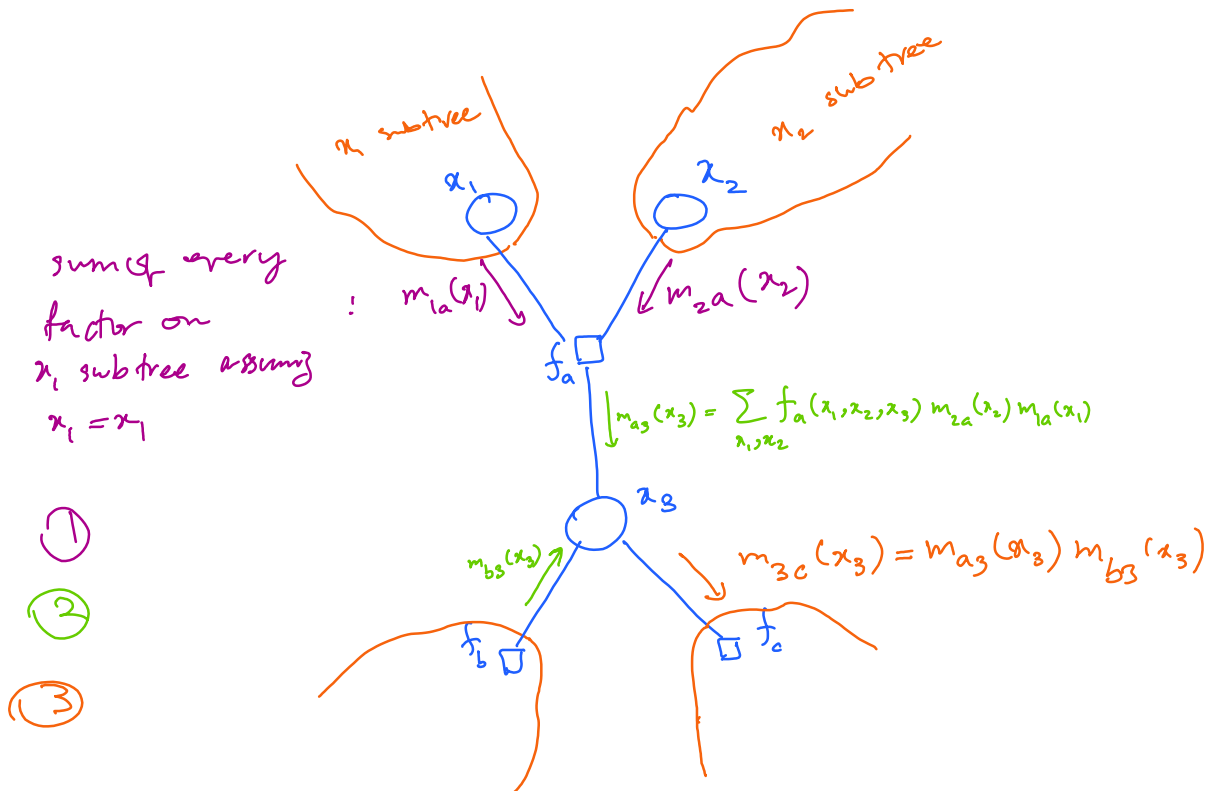
nodes  $\rightarrow$  variable nodes  
(CPDs at) nodes  $\rightarrow$  factor nodes



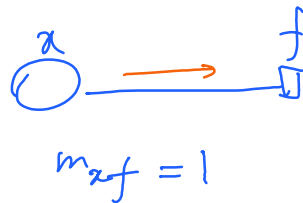
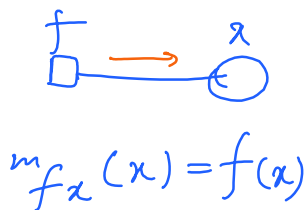
**Note:** even if the original MRF is not a tree, or the original BN has an MRF which is not a tree, the factor graph may still be a tree: that is discarding the types of the nodes in FG leads to a tree.

this is good because ↴

Sum-product for factor tree graphs:



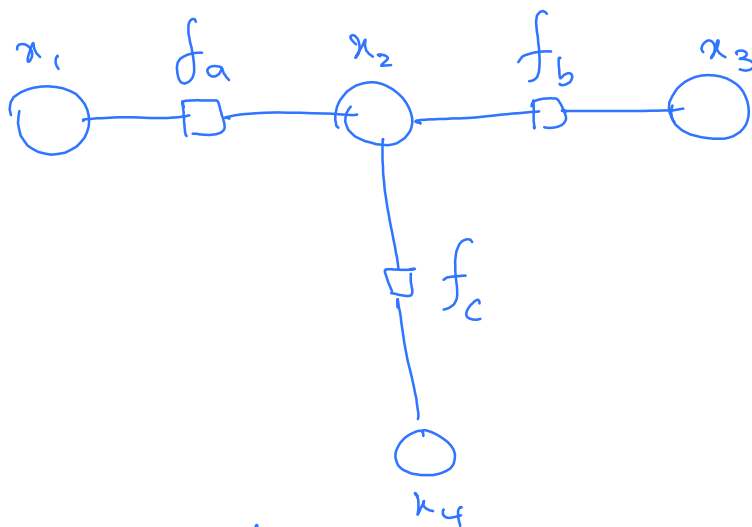
Starting steps at leaves:



Marginal at each node: product of msgs it receives

marginals at all nodes takes twice as much of that of a single node : two msgs per link as opposed to one.

Example:



round 1

round 2

$$\mu_{1a}(x_1) = 1$$

$$\mu_{4c}(x_4) = 1$$

$$\mu_{3b}(x_3) = 1$$

$$\mu_{a2}(x_2) = \sum_{x_1} f_a(x_1, x_2) \mu_{1a}(x_1)$$

$$\mu_{b2}(x_2) = \sum_{x_3} f_b(x_3, x_2) \mu_{3b}(x_3)$$

$$\mu_{c2}(x_2) = \sum_{x_4} f_c(x_2, x_4) \mu_{4c}(x_4)$$

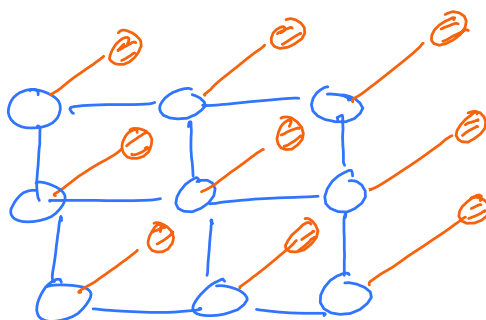
Problem: Identify the most likely configuration:

Find a set of values  $x_1^*, \dots, x_m^*$  s.t.

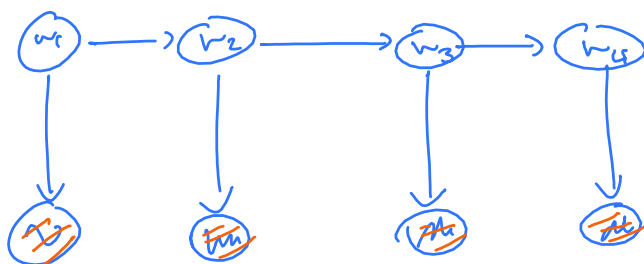
$$P(x_1^*, \dots, x_m^*) \geq P(x_i^m) \text{ for all } x_i^m$$

Applications:

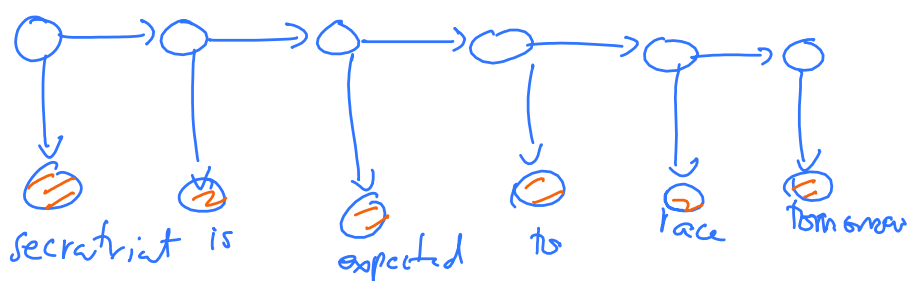
- Image denoising (Lab 1)



- Voice recognition



- Part of speech tagging



Do we already know how to solve this?

Finding most probable state for a node:

run sum-product and find  
state with max prob.

Finding most probable configuration for graph:

max-product

$$x_{\max} = \arg \max_x p(x)$$

May not be the same.

|       | $x=0$ | $x=1$ |
|-------|-------|-------|
| $y=0$ | 0.3   | 0.4   |
| $y=1$ | 0.3   | 0     |

$x=0$  max for  $x$

$y=0$  max for  $y$

$(x,y)=(0,1)$  max for  $(x,y)$

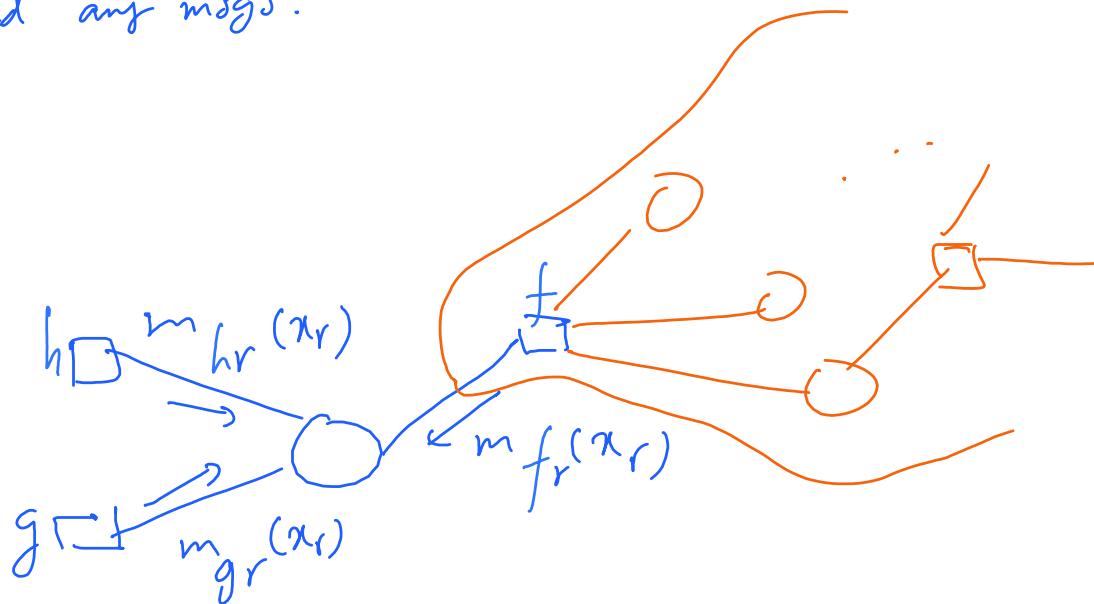
Let's instead try to find  $\max_{\vec{x}} p(\vec{x})$

$$\max_{x_1} p(\vec{x}_1) = \max_{x_1} \max_{x_2} \dots \max_{x_m} p(\vec{x}_1)$$

\* We can use a similar approach to elimination  
except that  $\Sigma$  is replaced with max.

\* Similarly sum-product can be turned  
to max-product to find  $\max p(x)$

\* Here, we pick an arbitrary root, which does not send any msgs.



The message from  $f$   $\mu_{fr}(x_r)$

Given the particular value for  $x_r$

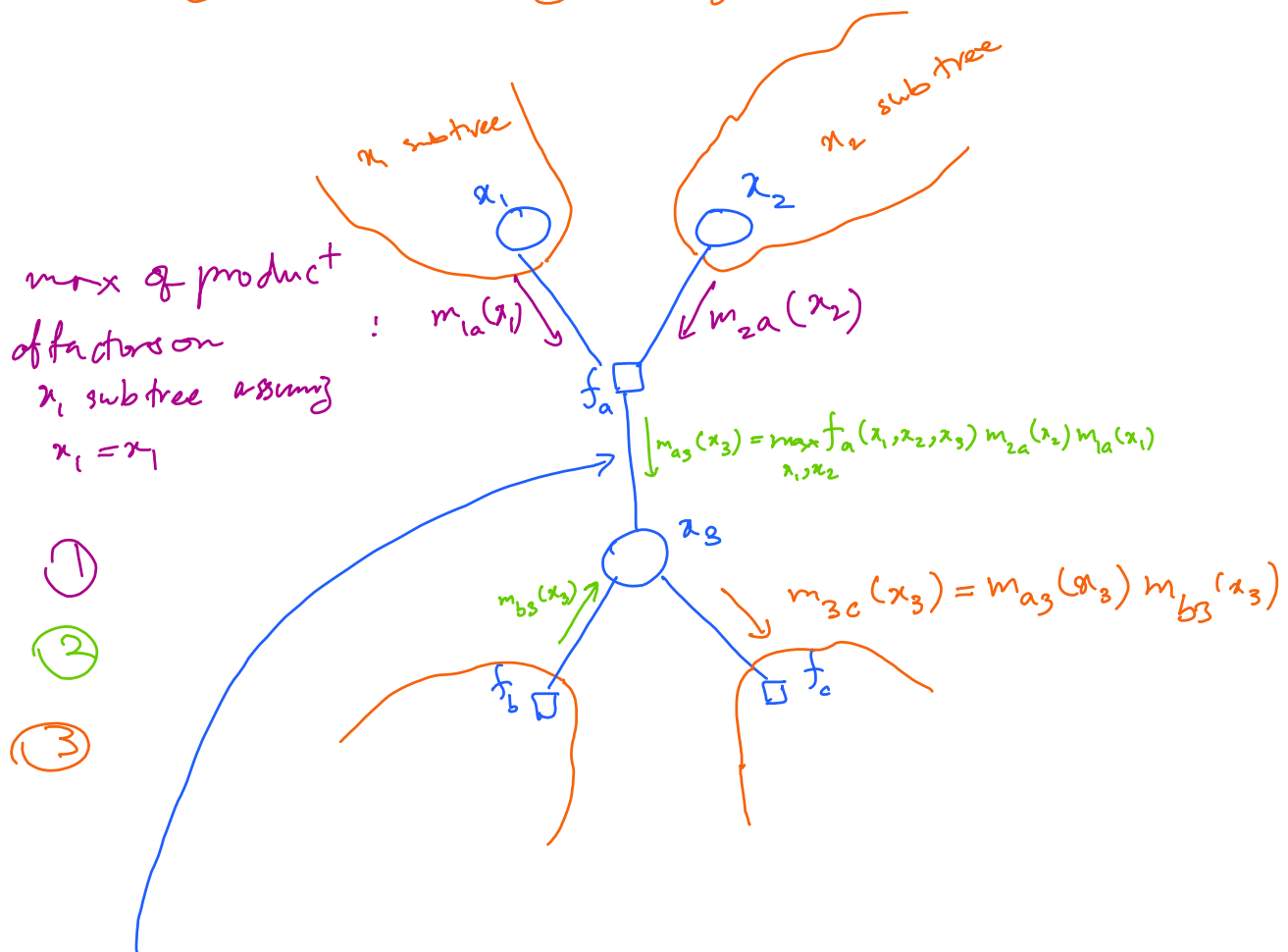
\* What is the most likely configuration for the subtree of  $f$

\* What is the "probability" of this configuration

$$\max p(\alpha_i^m) = \max_{x_r} \mu_{fr}(x_r) \mu_{gr}(x_r) \mu_{hr}(x_r)$$

from the value of  $x_r$  that maximizes this sum and messages we find the most likely configuration

\* Finding the maximizing configuration:



For each value of  $x_3$ , we also record which values of  $x_1, x_2$  achieved the max.

At the root, we find  $x_r^*$  that achieves the max. Then we back-track and find maximizing values for all nodes.

\* It may be more convenient to maximize

$\ln p(x) \Rightarrow$  max-sum algorithm

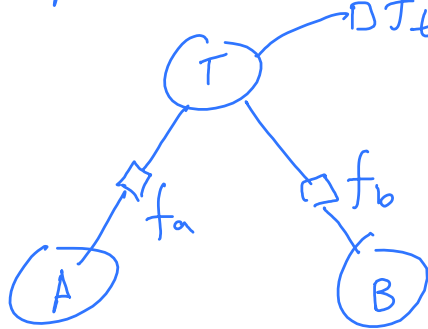
$$\sum f_i(x_{f_i})$$



$$\max_y p(x) = \max_{x_1 \dots x_n} \dots$$

note: why not continue the alg so that we can find  $x_i^*$  for all nodes just like how we found it for  $x_r$ ? We will find maximizing values, but they may belong to different maximizing configurations.

Example: most-likely configuration



$$f_t(T=0) = 0.65$$

$$f_t(T=1) = 0.35$$

$$f_a(A=0, T=0) = 0.9$$

$$f_a(A=0, T=1) = 0.5$$

$$f_a(A=1, T=0) = 0.1$$

$$f_a(A=1, T=1) = 0.5$$

$$f_b(B=0, T=0) = 0.82$$

$$f_b(B=0, T=1) = 0.15$$

$$f_b(B=1, T=0) = 0.18$$

$$f_a(B=1, T=0) = 0.12$$

$$\mu_{A^n}: \max \quad |$$

$$A=0 \longrightarrow |$$

$$A=1 \longrightarrow |$$

$$\mu_{aT} : \max_A f_a(A, T)$$

$$T=0 \longrightarrow 0.9 \text{ for } A=0$$

$$T=1 \longrightarrow 0.5 \text{ for } A=0 \text{ \& } A=1$$

$$\mu_{tT} : \max_T f_t(T)$$

$$T=0 \longrightarrow 0.65$$

$$T=1 \longrightarrow 0.35$$

$$\mu_{Tb} : \max_T \mu_{tT}(T) \mu_{a\bar{1}}(T) f_b(B, T)$$

...